

Node Class Documentation

Source File: Node.h
Namespace: dsp
Class Header: `template <class T> class Node`

Overview

The *Node* class builds a generic doubly-linked list.

Fields

- **data** (content of the node)
 - **Type:** Generic type T.
- **prev** (previous node link)
 - **Type:** Generic *Node* pointer.
- **next** (next node link)
 - **Type:** Generic *Node* pointer.

Constructors

- **Node()** (default constructor)
 - **Purpose:** Assigns the default value of the generic type to *data*, and assigns null to both *next* and *prev*.
- **Node(const T& value)**
 - **Purpose:** Assigns *value* to *data*, and assigns null to both *next* and *prev*.
 - **Parameter(s):**
 - *value*: The content of the node.

Non-Member Functions

- **Clone(Node<T>* rt)**
 - **Purpose:** Creates a deep copy of a null-terminated doubly-linked list.
 - **Parameters:**
 - *rt*: A null-terminated doubly-linked list.
 - **Return:** A null-terminated doubly-linked list copy.
- **Erase(Node<T>*& rt)**
 - **Purpose:** Deallocates a null-terminated doubly-linked list.
 - **Parameters:**
 - *rt*: Reference of a null-terminated doubly linked list.